DATA SCIENCE • MODULE 10

Data Science Projects

Build a Portfolio That Lands Your Dream DS Role

■ END-TO-END DS PROJECTS

■ LEARNING OUTCOME

By the end of this PDF you will have a complete framework for executing end-to-end data science projects, 5 ready-to-build project blueprints across key domains, and a portfolio strategy that makes you stand out in interviews.

■ Skills You Will Gain
• Execute an end-to-end DS project from problem to deployment
• Complete 5 domain-specific projects with real datasets
• Write professional Jupyter notebooks with documentation
• Build and share ML model demos with Streamlit
• Structure a DS portfolio that gets interview calls
• Answer technical DS interview questions confidently

■ **Hiring managers don't read resumes — they look at GitHub. 3 well-executed DS projects with deployed demos are worth more than any certification.**

SECTION 1

The DS Project Framework

1	**Problem Statement** Define: What question? What decision? Who benefits? What metric defines success?
2	**Data Acquisition** Kaggle, APIs, web scraping, company data — document source and license
3	**EDA & Cleaning** Profile, visualize, clean, handle nulls/outliers — minimum 10 charts
4	**Feature Engineering** Create, encode, scale, select — justify every choice
5	**Modeling** Try 3+ models, tune with GridSearchCV, compare by right metric
6	**Evaluation** Confusion matrix, ROC, feature importance, SHAP explanations
7	**Deployment** Streamlit app OR FastAPI endpoint OR Power BI dashboard
8	**Documentation** README: problem → approach → results → how to run → limitations

SECTION 2

Project 01 — Customer Churn Prediction

	Details
Domain	Telecom / SaaS
Dataset	Telco Customer Churn (Kaggle) — 7043 customers, 21 features
Problem	Predict which customers will cancel their subscription next month
Target Variable	Churn: Yes/No (binary classification)
Key Features	Tenure, MonthlyCharges, Contract type, InternetService, TechSupport
Models to Try	Logistic Regression, Random Forest, XGBoost — compare F1 + AUC
Expected AUC	~0.84–0.87 with proper feature engineering
Business Output	Churn probability score per customer + top 3 risk factors (SHAP)

Deployment	Streamlit app: input customer details → get churn probability + explanation

```
# Key analysis steps df['Churn'] = (df['Churn']=='Yes').astype(int) # encode target # Churn rate by contract type df.groupby('Contract')['Churn'].mean().plot(kind='bar', color='#7C3AED') # Feature importance after training importances = pd.Series(rf.feature_importances_, index=X.columns) importances.nlargest(10).plot(kind='barh') # SHAP explanation import shap explainer = shap.TreeExplainer(xgb_model) shap_values = explainer.shap_values(X_test) shap.summary_plot(shap_values, X_test)
```

SECTION 3

Project 02 — House Price Prediction

	Details
Domain	Real Estate / Finance
Dataset	Ames Housing (Kaggle) — 1460 houses, 79 features
Problem	Predict sale price of residential homes
Target Variable	SalePrice — continuous (regression)
Key Challenges	79 features, many nulls, highly skewed price distribution
Feature Eng.	Log-transform SalePrice, create TotalSF, HouseAge, QualityScore
Models to Try	Linear+Ridge+Lasso, Random Forest, XGBoost, Stacking
Expected RMSE	~$18K-$22K on validation — check Kaggle leaderboard
Deployment	Streamlit slider app: adjust rooms, location, size → see predicted price

SECTION 4

Project 03 — NLP Sentiment Analysis

	Details
Domain	NLP / E-commerce / Social Media
Dataset	Amazon Product Reviews or Twitter Sentiment (Kaggle)
Problem	Classify review sentiment: Positive / Negative / Neutral
Pipeline	Text cleaning → TF-IDF → Logistic Regression (baseline)
Advanced	BERT / DistilBERT fine-tuning with HuggingFace Transformers
Metrics	Accuracy + F1 (macro) — report per-class performance
Business Value	Auto-tag 10,000s of reviews, surface product issues instantly
Deployment	Streamlit app: paste any review → get sentiment + confidence score

```
from sklearn.feature_extraction.text import TfidfVectorizer from sklearn.linear_model import LogisticRegression from sklearn.pipeline import Pipeline import re def clean_text(text): text = text.lower() text = re.sub(r'[^a-z\s]', '', text) # remove punctuation return text df['clean'] = df['review'].apply(clean_text) # Build pipeline pipe = Pipeline([ ('tfidf', TfidfVectorizer(max_features=10000, ngram_range=(1,2))), ('clf',
```

```
LogisticRegression(max_iter=1000)) ]) pipe.fit(X_train, y_train) print(f'Test F1:
{f1_score(y_test, pipe.predict(X_test), average="macro"):.4f}')
```

SECTION 5

Project 04 — Time Series Forecasting

	Details
Domain	Retail / Supply Chain / Finance
Dataset	Rossmann Store Sales or M5 Forecasting (Kaggle)
Problem	Forecast daily sales for next 6 weeks
Baseline Models	Naive (last value), Moving Average, Exponential Smoothing
Advanced Models	ARIMA, SARIMA, Facebook Prophet, LightGBM with lag features
Feature Engineering	Lag features, rolling mean, day-of-week, holiday flags, promotions
Metric	RMSE + MAPE — interpretable as '±X%' error
Deployment	Dashboard showing forecast vs actuals with confidence intervals

```
from prophet import Prophet import pandas as pd # Prophet expects ds (date) and y (value) columns
df_p = df.rename(columns={'Date':'ds','Sales':'y'}) model = Prophet(yearly_seasonality=True,
weekly_seasonality=True, daily_seasonality=False) model.fit(df_p) future =
model.make_future_dataframe(periods=42) # 6 weeks forecast = model.predict(future)
model.plot(forecast) model.plot_components(forecast) # trend + seasonality
```

SECTION 6

Project 05 — Image Classification (Bonus)

	Details
Domain	Computer Vision / Healthcare / Manufacturing
Dataset	MNIST (digits), CIFAR-10, or Chest X-Ray Pneumonia (Kaggle)
Problem	Classify images into categories (10 digit classes or Normal/Pneumonia)
Baseline	Flatten image → Logistic Regression (weak, shows need for CNN)
Advanced	CNN with TensorFlow/Keras: Conv2D → MaxPool → Dense → Softmax
Transfer Learning	Use pretrained VGG16/ResNet50 — fine-tune last layers (fast, powerful)
Expected Accuracy	MNIST: ~99%, X-ray: ~92% with CNN
Deployment	Streamlit: upload image → get classification + confidence %

```
import tensorflow as tf from tensorflow.keras import layers, models # Simple CNN model =
models.Sequential([ layers.Conv2D(32,(3,3),activation='relu',input_shape=(28,28,1)),
layers.MaxPooling2D(2,2), layers.Conv2D(64,(3,3),activation='relu'), layers.MaxPooling2D(2,2),
layers.Flatten(), layers.Dense(128,activation='relu'), layers.Dropout(0.3),
layers.Dense(10,activation='softmax') ])
model.compile(optimizer='adam',loss='sparse_categorical_crossentropy',metrics=['accuracy'])
model.fit(X_train,y_train,epochs=10,validation_split=0.2)
```

SECTION 7

Deploying with Streamlit

```
# app.py — deploy any DS model in minutes pip install streamlit import streamlit as st import
pickle import pandas as pd st.title('Customer Churn Predictor') st.sidebar.header('Customer
Details') tenure = st.sidebar.slider('Tenure (months)', 0, 72, 12) monthly =
st.sidebar.number_input('Monthly Charges', 18.0, 120.0, 65.0) contract =
st.sidebar.selectbox('Contract', ['Month-to-month','One year','Two year']) # Load model model =
pickle.load(open('churn_model.pkl','rb')) if st.button('Predict Churn Probability'): input_df =
pd.DataFrame([[tenure,monthly,contract]], columns=['tenure','MonthlyCharges','Contract']) prob =
model.predict_proba(input_df)[0][1] st.metric('Churn Probability', f'{prob:.1%}') if prob > 0.5:
st.error('■■ HIGH RISK — Trigger retention campaign') else: st.success('■ LOW RISK') # Run:
streamlit run app.py
```

SECTION 8

DS Interview Questions

Question	Model Answer Structure
Walk me through an end-to-end DS project	Problem → Data → EDA → FE → Model → Eval → Deploy. Include specific numbers and
Why did you choose XGBoost over RandomForest?	XGBoost is boosting (sequential correction) vs bagging (parallel). On this tabular dataset X
How did you handle class imbalance?	'Detected in EDA: 85/15 split. Used class_weight=balanced + SMOTE + evaluated with F1
Explain a model's prediction to a non-technical stakeholder	'SHAP shows this customer churned because their monthly charge was high AND they're c
What would you improve with more time?	Shows maturity: 'Collect more data, add external signals, try deep tabular models, deploy w

■ BONUS RESOURCES

Kaggle	kaggle.com — datasets, notebooks, competitions with real prizes
Streamlit	streamlit.io — deploy ML apps in Python, free hosting
HuggingFace	huggingface.co — pre-trained models for NLP and vision
FastAPI	fastapi.tiangolo.com — serve ML models as REST APIs
MLflow	mlflow.org — experiment tracking and model registry
DVC	dvc.org — version control for datasets and models
Papers With Code	paperswithcode.com — SOTA models with code
DS Interview Prep	interviewquery.com, stratascratch.com

■ DS Projects Cheat Sheet	
Framework	Problem→EDA→FE→Model→Eval→Deploy→Docs
Churn dataset	Kaggle: Telco Customer Churn
House price	Kaggle: Ames Housing Dataset
NLP sentiment	Amazon Reviews / IMDB dataset
Time series	Kaggle: Rossmann / M5 Forecasting

Image class.	Kaggle: Chest X-Ray (Pneumonia)
pickle.dump(model,f)	Save trained model
pickle.load(f)	Load model for deployment
streamlit run app.py	Launch web app locally
shap.summary_plot	Explain model predictions
README must-haves	Problem+Data+Results+Demo+Code
Portfolio: 3-5 projects	Quality beats quantity always